Class 2, Week 9: Binary classification (yes or no) and how to measure it

Lecturer guide for CML3112, CML3113 and CML3111. Keep it private.

The class in one paragraph

Can a binary classifier say yes or no to "high cooling load", how does it learn, and how do we count and price its mistakes?

Welcome to Class 2. Last week the model gave you a number. Today it answers yes or no, which is called binary classification. We stay with the building data, and the question is whether the cooling load reaches its limit. You will see how a classifier is built, how it learns from its errors, and the most important lesson of the day: how to count its mistakes and put a price on them. The ideas come first, then the lab, then the quiz.

When each cohort meets

Cohort	Programmes	Date	Time and room
CML3112	Civil Engineering and Architecture	Tuesday 10 November 2026	11:00 to 14:00, Room 106
CML3113	Mining and Petroleum Engineering	Friday 13 November 2026	08:00 to 11:00, Room 105
CML3111	Electrical, Mechatronics, Robotics and Communications	Friday 13 November 2026	11:00 to 14:00, Room 115

Before the class

- Open the deck in Slide Show mode and the lab notebook in Colab before anyone arrives. Run the notebook from the top once on your account.
- Put the lab notebook and engineering_data.csv in the shared class folder and post the link. Every learner needs both files in Colab, in the Files panel.
- Write the "check your numbers" table on the board before the lab. Use your own cohort's column of the table "The numbers each cohort should see" below: the numbers differ between cohorts.
- The Block 1 quiz is online: forms Q7 (Class 1) and Q8 (Class 2). Open both in the last 20 minutes of the class. Each stays open for 24 hours (arrangement note of 2 November). Nothing is sat on paper.
- The deck states the Block 1 close from the arrangement note: CML3112 Tuesday 17 November, 11:00; CML3113 Friday 20 November, 08:00; CML3111 Friday 20 November, 11:00 (Kampala time). Confirm it in writing after the class.

The clock

Tuesday cohort (CML3112): the first full teaching of this class. Friday cohorts (CML3113, CML3111): the same deck and the same clock as the Tuesday cohort, because the concepts of this class are new to all three cohorts.

CML3112, Tuesday

Starts	Minutes	What happens
0:00	10	Recall from Class 1
0:10	40	Concepts 1: labels, spread, how a classifier learns
0:50	35	Concepts 2: counting and pricing mistakes
1:25	15	Break
1:40	60	Lab: binary classification, in the notebook

Starts	Minutes	What happens
2:40	15	Quiz opens: forms Q7 and Q8, 24 hours
2:55	5	Close and what is assessed

CML3113 and CML3111, Friday

Starts	Minutes	What happens
0:00	10	Recall from Class 1
0:10	40	Concepts 1: labels, spread, how a classifier learns
0:50	35	Concepts 2: counting and pricing mistakes
1:25	15	Break
1:40	60	Lab: binary classification, in the notebook
2:40	15	Quiz opens: forms Q7 and Q8, 24 hours
2:55	5	Close and what is assessed

Slide map

The deck has two halves. Slides up to the divider are concepts: no laptops. After the divider come the lab slides and the notebook. You switch once. The titles below are from the CML3112 deck. The other two decks have the same slides in the same order; where a title carries a number, their own number replaces it (see the per-cohort table).

Slide	Title	Part	Friday recap
1	Binary classification (yes or no) and how to measure it	Concepts	
2	Today in one view: what you will leave with	Concepts	
3	Your three hours: ideas first, then the lab, then the quiz	Concepts	
4	Three questions from last time	Concepts	
5	Answers to the three questions from Class 1	Concepts	
6	Words we use today	Concepts	
7	Binary classification gives a yes or no answer that tells a plant operator whether to act	Concepts	
8	A limit on the cooling load turns each row into a positive or negative label	Concepts	
9	The majority-class baseline always answers no, is right on 80 percent of rows and finds no actual positive	Concepts	
10	The standard deviation measures how far the values of a column typically sit from their mean	Concepts	
11	The standard deviation and the z-score as formulas, with training row 1 worked by hand	Concepts	
12	Logistic regression passes a linear score through the sigmoid (logistic) function, which stays between 0 and 1	Concepts	
13	Logistic regression in three lines: score, probability, prediction	Concepts	
14	Worked example: validation row 3 gets p = 0.45, so the model predicts no	Concepts	
15	Log loss (binary cross-entropy) measures how wrong one predicted probability is	Concepts	

Slide	Title	Part	Friday recap
16	The training loss is the mean log loss, and gradient descent walks the weights downhill	Concepts	
17	scikit-learn minimises the same log loss with a faster solver and a small L2 penalty that keeps the weights small	Concepts	
18	A decision tree classifier can also answer yes or no, and it does not need standardisation	Concepts	
19	The confusion matrix counts true and false positives and negatives	Concepts	
20	Accuracy and precision are worked out from the four counts TP, FP, FN and TN	Concepts	
21	Recall (sensitivity) and F1: how many actual positives we find, and one blended score	Concepts	
22	A false negative and a false positive cost different things for a building	Concepts	
23	Logistic regression and the decision tree give different counts on the same 15 rows	Concepts	
24	Lowering the decision threshold never lowers recall, and it can add false positives	Concepts	
25	The cost of the mistakes puts a price on each one: here a false negative costs 5 and a false positive costs 1	Concepts	
26	The cost table picks decision threshold 0.3, the lowest cost on the validation set	Concepts	
27	On a larger set, a lower decision threshold finds more positives and also gives more false positives	Concepts	
28	Two rates give the points of the ROC curve: the false positive rate (FPR) and the true positive rate (TPR)	Concepts	
29	The ROC curve plots one point for every decision threshold: FPR across, TPR up	Concepts	
30	AUC is the area under the ROC curve, and on a large dataset the curve is smooth	Concepts	
31	With 15 rows and 3 actual positives, every score here is shaky	Concepts	
32	Summary: judge a binary classifier by counting each kind of mistake and pricing it	Concepts	
33	Read the workflow before you write it: impute, standardise, fit, predict probabilities, count	Concepts	
34	Part 2: the lab	Lab	
35	The lab has nine steps in five stages, and each stage ends with a check	Lab	
36	Check your numbers against these before you move on	Lab	
37	Four messages you will meet, what they mean and what to do	Lab	
38	How this class counts towards your coursework	Lab	
39	Save the notebook and keep your cost table	Lab	

Questions to ask, and a good answer

Ask	A good answer
What does a false positive cost for your equipment?	Answers differ. Listen for a cost in time or energy, and compare it with the cost of a false negative.
Why is accuracy alone not enough?	With few positive cases, the majority-class baseline scores a high accuracy and finds none of them.

Ask	A good answer
When is precision the more important score?	When a false positive is expensive, for example an unnecessary stop.
When is recall the more important score?	When a false negative is dangerous or expensive, for example an overheating motor.
What happens to recall if you lower the decision threshold?	It rises or stays the same, never falls. The table gives recall at 0.3, 0.5 and 0.7 for each cohort.
What is the log loss of an actual positive row with p = 1?	Zero: -ln 1 = 0. The model was certain and right.
Why do the weights change without StandardScaler?	Each weight now belongs to one unit of its feature, and the L2 penalty adds the squared weights in those units, so the fitted model is different.
Why do we fit StandardScaler on the training set only?	If the validation set helped decide the mean and the standard deviation, information would leak from held-out data into the model. That is data leakage.
Does the decision tree need standardisation?	No. A tree only asks whether a value is above or below a threshold, and standardisation does not change the order of the values.
What does an AUC of 0.5 mean?	The model ranks positive and negative cases no better than a coin flip.

Ideas learners often get wrong

What a learner may say	What to say back
High accuracy means a good model	On the validation set the majority-class baseline (always predict no) already scores the accuracy in the table below. Compare with that baseline and look at recall.
The decision threshold is always 0.5	0.5 is a default. Choose the threshold from the cost table: the lowest validation cost, and a tie goes to the higher threshold.
Standardisation cannot change the answer of the model	For a tree that is right. For LogisticRegression it can: scikit-learn adds an L2 penalty on the weights (C = 1.0), and the penalty depends on the units. Without StandardScaler the weights change, the probabilities move, and in some cohorts the confusion matrix at 0.5 changes (see the table).
Log loss and accuracy measure the same thing	Accuracy only counts right and wrong. Log loss also punishes a confident wrong probability: -ln 0.01 is 4.6, while -ln 0.5 is 0.69.
An AUC of 1.0 means the model is perfect	It means every actual positive scored above every actual negative in these 15 rows. New equipment can behave differently.

Walking the lab

Step	What to look for
Step 1: prepare	Rows after cleaning 88. Split 58, 15, 15. The same cleaning and split as Class 1, repeated.
Step 2: label	Positive labels and the baseline accuracy: see the table below (they differ by cohort).
Step 3: logistic regression	Weights, intercept and the mean training log loss as in the table. The CHANGE task removes StandardScaler: the load weight becomes much smaller, because it now belongs to one percent of load, and the fitted model changes because of the default L2 penalty. Ask which weight shrank, which grew, and whether the confusion matrix changed (the table has the answer per cohort). No ConvergenceWarning appears with max_iter=1000.
Step 4: decision tree	Depth 3. TP, FN, FP, TN as in the table. Some leaves rest on one training row: point at the numbers in brackets on the tree slide.

Step	What to look for
Step 5: confusion matrix and scores	Counts and scores as in the table. Ask one learner to repeat one division at the board.
Step 6: cost table and threshold	Costs at 0.3, 0.5 and 0.7 and the chosen threshold as in the table (cost = 5 x FN + 1 x FP, invented teaching costs). The CHANGE task adds 0.1 and 0.2: false positives appear in every cohort.
Step 7: ROC curve and AUC	AUC as in the table. The plot shows only the corner points that roc_curve keeps (the number is in the table). Ask what a score this high on 15 rows promises.
Steps 8 and 9: conclusion and save	Three sentences from every learner. Insist on restart and run all before saving.

The numbers each cohort should see

Check	CML3112 (kW)	CML3113 (kWh/t)	CML3111 (deg C)
Label limit	54 kW	13.5 kWh/t	80 deg C
Standard deviation load_pct, ambient_c: StandardScaler (divide by n); pandas .std() (n - 1)	22.73, 4.55; 22.93, 4.59	22.73, 4.55; 22.93, 4.59	22.73, 4.55; 22.93, 4.59
Positive labels, training set	19 of 58	24 of 58	17 of 58
Positive labels, validation set	3 of 15	3 of 15	3 of 15
Majority-class baseline accuracy	0.80	0.80	0.80
Logistic weights: load, ambient, intercept	2.79, 0.73, -1.71	2.64, 0.49, -0.95	2.81, 0.93, -1.99
Logistic: TP, FN, FP, TN	2, 1, 0, 12	3, 0, 1, 11	2, 1, 0, 12
Logistic: accuracy, precision, recall, F1	0.93, 1.00, 0.67, 0.80	0.93, 0.75, 1.00, 0.86	0.93, 1.00, 0.67, 0.80
Decision tree depth 3: TP, FN, FP, TN	2, 1, 1, 11	3, 0, 1, 11	2, 1, 0, 12
Decision tree depth 3: accuracy, precision, recall, F1	0.87, 0.67, 0.67, 0.67	0.93, 0.75, 1.00, 0.86	0.93, 1.00, 0.67, 0.80
Recall at decision thresholds 0.3, 0.5, 0.7	1.00, 0.67, 0.00	1.00, 1.00, 0.67	1.00, 0.67, 0.00
FP at decision thresholds 0.3, 0.5, 0.7	1, 0, 0	2, 1, 0	0, 0, 0
AUC	1.00	0.97	1.00
Mean training log loss	0.21	0.25	0.19
Log loss by hand (p rounded to 2 decimals): validation row 3; the actual negative with the highest p	0.80; row 11, 0.49	0.63; row 11, 0.92	0.76; row 11, 0.36
Gradient-descent slopes at the start (b0, b1, b2)	0.17, -0.34, -0.04	0.09, -0.38, 0.00	0.21, -0.31, -0.08
Cost (5 FN + 1 FP) at 0.3, 0.5, 0.7; chosen threshold	1, 5, 15; 0.3	2, 1, 5; 0.5	0, 5, 15; 0.3
No StandardScaler (Step 3 CHANGE): weights load, ambient	0.547, 0.841	0.326, 0.481	0.765, 1.239
No StandardScaler: TP, FN, FP, TN at 0.5; largest probability change	3, 0, 0, 12; 0.35	3, 0, 1, 11; 0.34	3, 0, 0, 12; 0.41

Check	CML3112 (kW)	CML3113 (kWh/t)	CML3111 (deg C)
FP at thresholds 0.1 and 0.2 (Step 6 CHANGE)	2, 2	3, 2	2, 2
Points in the student ROC plot (roc_curve default)	4	6	4

If things go wrong

- If the concepts run over, skip the illustrative larger-set table and the smooth AUC slide, and say each in one sentence. Keep the standard deviation, log loss, gradient descent, confusion matrix and cost table slides.
- If only 40 minutes of lab remain, do steps 1 to 3, 5 and the cost table in step 6. Set the decision tree and the ROC curve as practice for next time.
- If the quiz time is under pressure, shorten the lab by dropping the decision tree. The quiz forms must still open today.
- If wifi fails in the room, use the offline set-up from Week 2: the notebook runs unchanged in VS Code.
- If several learners are stuck at the same cell, stop the room, run that cell on the projector and let them copy the fix.

After the class

- Share the lab notebook and engineering_data.csv again so absent learners can follow.
- Confirm in writing the Block 1 close stated on the deck and the two quiz forms.
- Note any question you could not answer and start Class 3 with it.

Speaker script

Every slide carries its spoken script in the PowerPoint notes (open Presenter View). The scripts are first person and are meant to be read or paraphrased. Pauses are marked by questions.